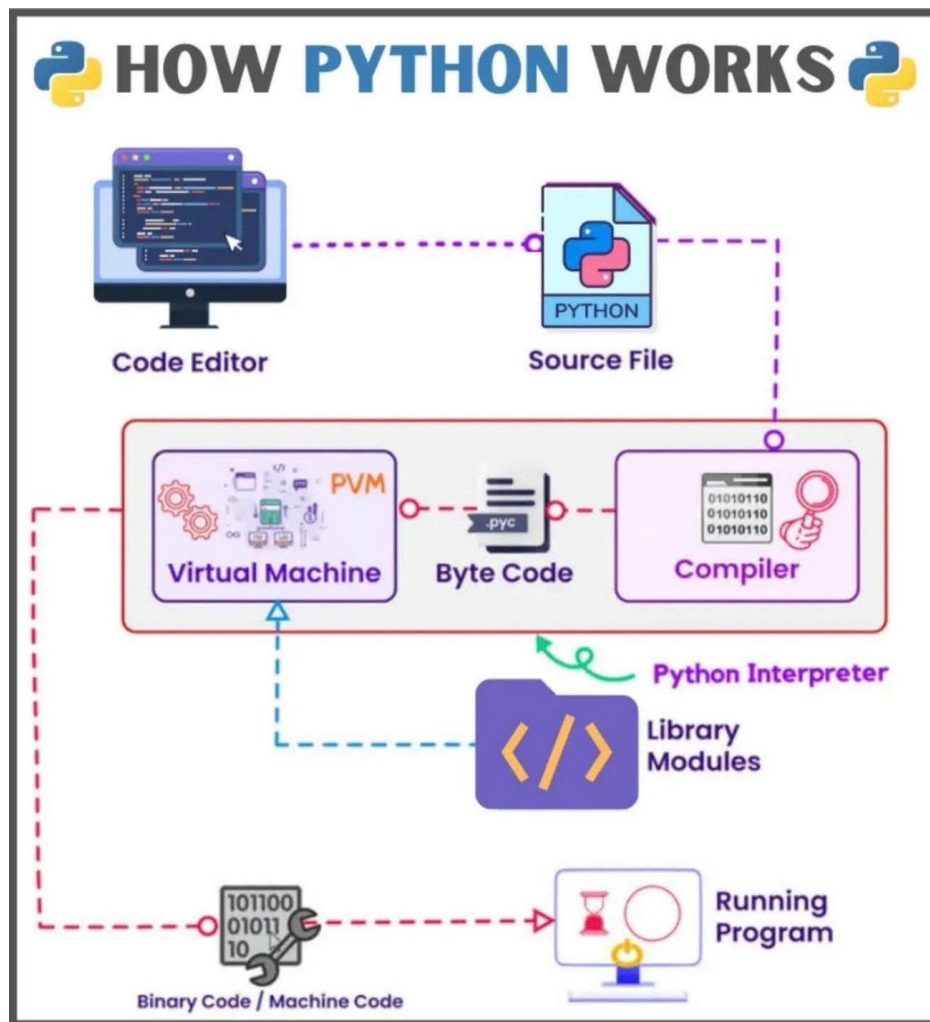




1. Writing the Code

- The programmer writes Python code using a **Code Editor** (VS Code, PyCharm, IDLE, etc.).
- The file is saved as a **Python Source File** (.py).

2. Python Interpreter Starts

- When we run the .py file, it is handled by the **Python Interpreter**.

- Python is called an **interpreted language**, but internally it also **compiles** code.
-

3. Compilation to Bytecode

- The **Python Compiler** checks:
 - Syntax errors
 - Code structure
- If no errors are found:
 - The source code is converted into **Bytecode**
 - Bytecode is stored as `.pyc` files (internally)

Bytecode is NOT machine code and NOT human-readable.

4. Python Virtual Machine (PVM)

- The **PVM (Python Virtual Machine)** takes the bytecode.
 - PVM executes the bytecode **line by line**.
 - This is why Python feels like an interpreted language.
-

5. Use of Libraries & Modules

- During execution, Python may use:
 - Built-in libraries
 - External modules

- These support program functionality.
-

6. Conversion to Machine Code

- Internally, bytecode is converted into **Binary / Machine Code**
 - This code is understood by the **CPU**
-

7. Program Output

- Finally, the **Running Program** produces output:
 - Console output
 - File operations
 - GUI or web responses
-

One-Line Explanation:

Python code → Bytecode → PVM → Machine Code → Output

“Python is an interpreted language that internally compiles source code into bytecode, which is executed by the Python Virtual Machine.”